

Supplementary Table S8 — Feature comparison: OligoVigil vs. existing oligonucleotide databases

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Scope and inclusion criteria

We compared OligoVigil against five existing curated resources that overlap in scope with oligonucleotide-therapeutic safety or activity annotation: theRNA (Zhou *et al.*, 2026), siRNAEfficacyDB (Zhang *et al.*, 2024), CMsiRNAdb (He *et al.*, 2026), siRNAmoD (Dar *et al.*, 2016) and CRISPROffT (Wang *et al.*, 2025). The first four catalogue therapeutic RNA or siRNA records directly comparable in molecule class to the OligoVigil release. CRISPROffT is included as a functionally adjacent reference for the off-target annotation column only; its target molecule class (Cas guide RNAs in editing constructs) differs from the antisense/siRNA chemistry in OligoVigil, and reviewers should weight its comparison cells with that caveat in mind. No further oligonucleotide-therapeutic safety databases meeting the NAR Database Issue access bar (open, citable, indexed) were identified during the verification sweep. All cells were verified against the cited primary source or the live portal on 2026-06-08; cells marked “—” reflect features the source did not document or that could not be independently verified on that date.

Table S8

Feature	OligoVigil	theRNA ¹	siRNAEfficacyDB ²	CMsiRNAdb ³	siRNAmoD ⁴	CRISPROffT ⁵
Total curated records	737 release rows (626 toxicity + 111 off-target)	6,860 records / 1,310 diseases / 4,161 publications	3,544 siRNA records (42 genes, 5 cell lines)	43,153 entries (15,760 unique sequences, 36 modifications, 13 genes)	4,894 modified siRNAs (128 modification types)	226,298 predicted / 8,940 experimentally validated off-targets (371 gRNAs)
Source-anchored provenance (1:1 quote + PMID + section)	1:1 verbatim quote + PMID + section, all 737 rows; W3C PROV chain	yes (PMID + journal + year per entry)	partial (publication list, no per-row verbatim quote)	partial (patent identifier per entry, no per-row quote)	partial (PMID per record, no quote/section)	partial (study + PubMed link, aggregate provenance)
Inter-rater reliability reported (κ or equivalent)	Cohen κ_{binary} = 0.42 (drop-abstain, n=92) / 0.34 (collapse-abstain, n=100) on 100-row dual-rater packet; A10 third-adjudicator pending	—	—	—	—	—

Feature	OligoVigil	theRNA ¹	siRNAEfficacyDB ²	CMsiRNAdb ³	siRNAmod ⁴	CRISPROffT ⁵
AI/ML pre-curation false-accept audit	FAR 0.73 (Wilson 95% CI [0.63, 0.81]) on machine pre-curation, disclosed	—	—	— (NLP-assisted curation, no audit reported)	—	—
Benchmark + deterministic baselines (frozen split, pair-level isolation)	344-row benchmark, deterministic prior baselines, pair-level isolation, 4.1% placeholder contamination disclosed	—	—	—	—	partial (29 prediction techniques compared, no frozen public split with deterministic baselines)
Structured assay metadata (dose, exposure, tissue, organism)	dose_value 0.3%, exposure_time_value 0%, tissue/organism uneven	partial (disease + modality fields, no dose/exposure schema)	yes (cell line + target gene + concentration captured)	partial (target gene + modification position, no dose/exposure schema)	— (modification metadata only)	partial (cell line/tissue field, no dose/exposure)
Chemical-modification coverage (per-position annotation)	— (no modification-position schema; only free-text quote)	partial (modality tag, no per-position schema)	—	yes (36 modification types, per-position annotation, 15,760 unique modified sequences)	yes (128 modification types, per-position annotation)	— (not applicable)
Off-target gene-level resolution	4/111 off-target rows at specific-gene resolution; remainder transcriptome-wide	—	—	—	—	yes (gene/transcript-level off-target pairs, 8,940 experimentally validated)
Open no-login web portal	yes (HTTPS, no login)	yes	yes	yes	— (portal http://crdd.osdd.net/servers/sirnamod/ unreachable 2026-06-08)	yes
Programmatic interfaces (REST / OpenAPI / MCP / Bioschemas)	REST + OpenAPI 3.1 + MCP server + Bioschemas JSON-LD + SQL view release_audit_v	partial (web search/download, no published API spec)	partial (web search/download)	partial (web search/download)	—	partial (web search/download, no published API spec)
Versioned bulk download + per-release DOI	yes (Zenodo snapshots, per-release DOI, CSV + JSON-LD)	partial (bulk download, no per-release DOI documented)	partial (bulk download, no per-release DOI documented)	partial (bulk download, no per-release DOI documented)	partial (single 2016 release, no DOI)	partial (bulk download, no per-release DOI documented)

Feature	OligoVigil	theRNA ¹	siRNAEfficacyDB ²	CMsiRNAdb ³	siRNAmo ⁴	CRISPROffT ⁵
License (code + data)	MIT (code) + CC BY 4.0 (data)	—	yes (CC BY-NC, declared on portal)	—	—	partial (paper CC BY-NC via OUP; portal carries no separate data/code license)
Maintenance commitment + named curator	5-year maintenance commitment; single named curator (Ni Jie, AI-assisted)	yes (Ren/Zuo labs, NAR 2026 release)	yes (Zhang <i>et al.</i> maintained release, 2024)	yes (He <i>et al.</i> maintained release, 2026)	partial (stated half-yearly/annual update commitment in 2016 paper; portal currently unreachable)	yes (Zhou lab UHealth, 2025 release)

¹ **theRNA** — URL: <https://therna.renlab.cn/>; Citation: Zhou Y, Hu Y, Zhang L, Wu T, Peng S, Lin C, Wang Z, Zhou Z, Zheng J, Ren J, Zuo Z. theRNA: a curated knowledgebase of functional RNA therapeutics spanning diverse modalities and disease applications. *Nucleic Acids Research*. 2026;54(D1):D1672–D1682. doi:10.1093/nar/gkaf1064; Records: 6,860 experimentally validated RNA therapeutics targeting 1,310 diseases, curated from 4,161 publications (screened 11,126), 2026.

² **siRNAEfficacyDB** — URL: <https://cellknowledge.com.cn/siRNAEfficacy>; Citation: Zhang Y *et al.* siRNAEfficacyDB: an experimentally supported small interfering RNA efficacy database. *IET Systems Biology*. 2024. doi:10.1049/syb2.12102; Records: 3,544 siRNA records (42 target genes, 5 cell lines), 2024.

³ **CMsiRNAdb** — URL: <https://cellknowledge.com.cn/CMsiRNAdb/>; Citation: He S, Chen C, Pan X *et al.* CMsiRNAdb: a database of chemically modified siRNA silencing efficiency for nucleic acid drug design. *BMC Bioinformatics*. 2026;27(1):33. doi:10.1186/s12859-025-06359-y; Records: 43,153 experimentally validated chemically modified siRNA entries (15,760 unique sequences), 36 modification types, 13 target genes, from 90 patents, 2026.

⁴ **siRNAmo** — URL: <http://crdd.osdd.net/servers/sirnamod/> (portal unreachable on 2026-06-08); Citation: Dar SA, Thakur A, Qureshi A, Kumar M. siRNAmo: a database of experimentally validated chemically modified siRNAs. *Scientific Reports*. 2016;6:20031. doi:10.1038/srep20031; Records: 4,894 chemically modified siRNA sequences (128 unique chemical modifications), 2016.

⁵ **CRISPROffT** — URL: <https://ccsm.uth.edu/CRISPROffT/>; Citation: Wang G, Liu X, Wang A, Wen J, Kim P, Song Q, Liu X, Zhou X. CRISPROffT: comprehensive database of CRISPR/Cas off-targets. *Nucleic Acids Research*. 2025;53(D1):D914–D924. doi:10.1093/nar/gkae1025; Records: 226,298 potential off-targets across 371 guide-RNA sequences; 8,940 experimentally validated; 74 studies; 85 Cas/gRNA combinations; 34 cell lines/tissues, 2025. Included as functionally adjacent reference (different molecule class).

⁶ **siRNAEfficacyDB and CMsiRNAdb co-hosting** — siRNAEfficacyDB and CMsiRNAdb are both hosted under the cellknowledge.com.cn institutional portal but were published independently (Zhang *et al.* 2024 vs He *et al.* 2026, different author groups); both portals were accessed and verified independently on 2026-06-08.

What OligoVigil uniquely provides

Five rows in Table S8 distinguish OligoVigil from every confirmed competitor on the verification date. (i) Inter-rater reliability is reported as a Cohen κ_{binary} of 0.42 (drop-abstain) or 0.34 (collapse-abstain), with a third-adjudicator packet (A10) prepared; no other DB in the comparison publishes an agreement statistic. (ii) The AI/ML pre-curation step is audited rather than asserted: a false-accept rate of 0.73 (Wilson 95% CI [0.63, 0.81]) is released alongside the data, whereas competitors that use NLP-assisted screening (notably theRNA's 11,126 $\rightarrow$ 4,161 filter, and CMsiRNAdb's patent extraction) do not quantify their machine-stage error. (iii) A frozen 344-row benchmark with deterministic prior baselines and pair-level train/test isolation is shipped with the release, including the disclosed 4.1% placeholder contamination — comparable in form to the model-evaluation panels in CRISPROffT but distinct in releasing the split and the baselines rather than only the comparison numbers. (iv) Programmatic surface is broader: REST plus OpenAPI 3.1 plus an MCP server plus Bioschemas JSON-LD plus the `release_audit_v` SQL view, against web-search-and-download portals elsewhere. (v) Per-release DOI through Zenodo with paired CSV and JSON-LD snapshots; the comparator portals offer bulk download but do not document per-version DOIs.

Where competitors lead OligoVigil

The principal weakness is absolute scale: OligoVigil’s 737 release rows are between one and two orders of magnitude smaller than theRNA (6,860), siRNAEfficacyDB (3,544), CMsiRNAdb (43,153) and siRNAmoD (4,894), and three orders smaller than CRISPRoffT (226,298 predicted / 8,940 validated). This is discussed under §Limitations of the main text. On chemical-modification coverage, both CMsiRNAdb (36 modification types, 15,760 unique sequences, per-position annotation) and siRNAmoD (128 modification types, per-position annotation) substantially exceed OligoVigil, which carries no modification-position schema and captures modifications only when the source quote happens to mention them. On structured assay metadata, siRNAEfficacyDB provides cell-line and concentration fields across its 3,544 records, against the 0.3% dose-value and 0% exposure-time-value coverage in OligoVigil — assay metadata is the weakest column in the OligoVigil release. On off-target gene-level resolution, CRISPRoffT pins 8,940 experimentally validated off-targets to specific transcripts/genes, against four specific-gene rows in OligoVigil’s 111 off-target entries; the remainder are transcriptome-wide aggregates and should be treated as coarse signals only. Of 78 data cells in Table S8 (13 features $\times$ 6 database columns), 29 are populated with “—” reflecting either an undocumented feature in the source or a feature that could not be verified on 2026-06-08; 28 of these are competitor cells and 1 is the OligoVigil chemical-modification-coverage cell.